Exam Date & Time: 31-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

VII Semester End Sem Make Up Examination Dec/Jan 2025

Tissue Engineering [BME 4071]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

1A)	Analyse the signal transduction and gene activation process pertaining to cell signalling in bone (discuss with respect to bone morphogenic protein (BMP).	(5)
1B)	Explain the significance of the presence of embryoid bodies in stem cell culture.	(2)
1C)	Illustrate briefly, the role of various transcription factors pertaining to the downregulated NANOG expression of mouse embryonic stem cells.	(3)
2A)	Analyse the limitation of snap freezing in the decellularization process.	(2)
2B)	Analyse how histone switch and chromatin compaction regulate epigenetic mechanism.	(4)
2C)	Analyse the interdependence of catenin and the transcription factors (SOX9 and RUN X2) in deciding the direction of differentiation. Highlight the significance of HOX gene cluster in bone tissue engineering.	(4)
3A)	Sourish wishes to fabricate interconnected homogeneous porous scaffolds. Highlight his strategies in a brief and logical manner (use schematics if necessary).	(5)
3B)	Compare the significance of Glycine-X-Y combination in the collagen fibril. Analyse whether this would have any impact on the isolation process of collagen.	(3)
3C)	Explain how image processing tool would avoid subjectivity while assessing porosity based on SEM micrographs of polymeric scaffolds.	(2)
4A)	Compare and analyse selection processes pertaining to gradient centrifugation process.	(4)
4B)	Explain with proper reason: (a)Disposable tissue culture vessels are made of polystyrene.	(3)

(b)Culture condition (2D/3D) influences cellular behaviour.

4C) Compare different types of tissue culture medium. (3)

5A) Compare simple, complex and storage niche. (3)

5B) How do you extract feeder cell layers? Explain its importance in co-culture and stem cell culture (be specific to analyse its role). (3)

5C) Compare selective adhesion and antibody panning methods of cell selection. (4)

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